# INVESTIGATIONS ON

# HYBRID PHASE CHANGE MATERIAL COMPRISING GRAPHENE NANOPARTICLE, COPPER NANOPARTICLE AND PARAFFIN WAX

by

***Owais Ur Rahman (22911A0307)***

***Mohammed Osama (22911A0305)***

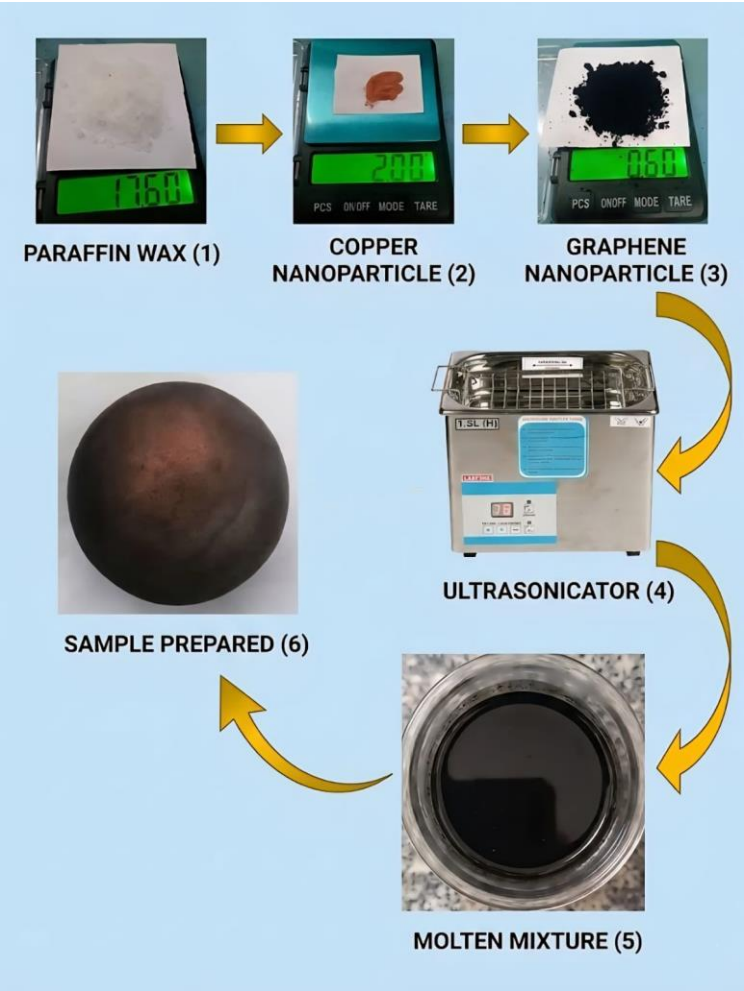
***Guide: Dr J. Emeema***

Phase Change Materials (PCMs) are widely used for thermal energy storage due to their ability to store and release latent heat during melting and solidification [1]. However, conventional PCMs such as paraffin wax suffer from very low thermal conductivity, which slows down heat absorption and release, making them less efficient for practical applications. To overcome this limitation, the present work focuses on developing a new nano-enhanced PCM (NEPCM) by incorporating copper nanoparticles and graphene nanoparticles into paraffin wax. Based on the analysis of previous research articles, this specific combination of PW, Cu and Graphene together in a single composite PCM has not been explored earlier, making the study novel.

Three composite PCM samples are prepared with fixed 10 wt% copper nanoparticles and varying 1%, 2%, and 3% graphene nanoparticles, while paraffin wax acts as the base PCM. The samples are fabricated using the ultrasonication method, which helps to break nanoparticle agglomerations and ensures uniform dispersion throughout the PCM matrix. After preparation, the samples undergo Scanning Electron Microscopy (SEM) to analyze surface morphology, clustering behaviour, and dispersion of nanoparticles. Further, thermal conductivity testing is carried out enabling comparison of pure paraffin with the nano-enhanced PCM samples.

Key words: paraffin wax, graphene nanoparticle and copper nanoparticle

## Graphical Representation




## Introduction

Thermal energy storage has become a critical requirement in modern engineering systems as industries increasingly move toward energy-efficient and environmentally sustainable technologies. Among the various energy storage methods, latent heat storage using Phase Change Materials (PCMs) is considered one of the most effective due to its ability to store large quantities of heat within a narrow temperature range. When a PCM undergoes a phase transition typically from solid to liquid or vice versa it absorbs or releases latent heat while maintaining an almost constant temperature. This unique property makes PCMs extremely valuable in applications that require temperature stability, thermal buffering, and passive thermal management.

PCMs have been successfully used in several fields such as building construction (thermal insulation panels, roofing systems), electronics cooling (battery packs, processors, heat-generating components), solar thermal storage (smoothening day–night energy availability), refrigerated transportation, medical cold-chain logistics, and wearable textiles. As global energy consumption continues to rise, the importance of effective thermal management

solutions has increased significantly, especially in systems that depend on renewable or intermittent energy sources. In such systems, PCMs can drastically reduce peak thermal loads, maintain safe operating temperatures, and enhance overall system efficiency.

Among commercially available PCMs, paraffin wax is one of the most popular choices. It offers several advantages including good thermal stability, chemical inertness, negligible supercooling, compatibility with long-term thermal cycling, and ease of manufacturing. Paraffin also has a wide range of melting temperatures based on its composition, allowing engineers to select a suitable grade for specific applications. Despite these advantages, paraffin wax suffers from a major drawback of extremely low thermal conductivity, generally between 0.2–0.4 W/mK. This significantly reduces the rate at which heat is absorbed or released, making the melting and solidification processes slow. As a result, its usage becomes limited in high-performance or fast-response applications.

To address this limitation, researchers have increasingly focused on enhancing the thermal conductivity of paraffin wax by adding thermally conductive materials [2]. Nanoparticles have emerged as one of the most promising solutions because even small quantities can drastically improve thermal behavior. Metal nanoparticles such as copper (Cu) provide excellent thermal conductivity and can act as efficient heat conduction pathways within the PCM matrix. On the other hand, graphene nanoparticles offer extraordinarily high thermal conductivity reported to be up to 5000 W/mK [3] along with excellent mechanical strength, chemical stability, and very large surface area. Both materials independently show potential to enhance paraffin PCM performance.

However, based on the review and analysis of previous research works, it is evident that very limited studies have explored the combined effect of copper nanoparticles and graphene nanoparticles together in a single PCM composite, especially in controlled and optimized weight ratios [4], [5], [2], [3]. Most available studies focus on either copper-enhanced PCM or graphene-enhanced PCM independently. The possibility of a synergistic effect, where the combination of both high-conductivity nanoparticles could produce even greater improvement than using them individually, has not been sufficiently examined. This gap in literature forms the core motivation for the present study.

In this project, a novel composite PCM is developed by incorporating 10 wt% copper nanoparticles and varying concentrations of 1%, 2%, and 3% graphene nanoparticles into paraffin wax. These specific percentages are chosen based on findings from earlier studies and the need to compare how increasing graphene concentration affects thermal behavior. The preparation of the composite PCM samples is carried out using the ultrasonication method, which is widely recognized as one of the most effective dispersion techniques.

Ultrasonication helps in breaking down nanoparticle agglomerations and ensures uniform distribution of nanoparticles throughout the paraffin matrix, which is crucial for consistent thermal performance.

After the fabrication of the samples, the project includes a detailed characterization phase. Scanning Electron Microscopy (SEM) is used to analyze surface morphology, observe the dispersion of copper and graphene nanoparticles, and identify any clustering or agglomeration. Additionally, thermal conductivity testing is conducted to quantitatively assess the improvement in heat transfer capability. These tests together help in evaluating whether the newly developed composite PCM demonstrates enhanced performance compared to pure paraffin wax.

This Stage-I report aims to establish the complete theoretical foundation, material selection rationale, preparation methodology, and testing plan required for the experimental study. By introducing a novel PCM formulation and outlining the experimental framework, this work sets the stage for Stage-II, where full performance evaluation, thermal cycling, and comparative analysis will be conducted. The overall goal is to create a high-performance PCM that can offer improved conductivity, faster thermal response, and enhanced applicability in modern thermal management systems.

**Literature Survey Table:**

| S. No. | References | Composite | wt./vol% | Thermal Conductivity (k) | Latent Heat / Thermal Performance |
|---|---|---|---|---|---|
| 1 | Chang et al. (2023) [1] | Paraffin + Copper–Graphene porous composite | Not specified | Significant enhancement in thermal conductivity due to conductive network | Improved thermal storage stability and reduced leakage |

| | | | | | |
|---|---|---|---|---|---|
| **2** | **Calotă et al. (2024) [2]** | **Paraffin + Hybrid nanoparticles (metal + graphene)** | **Various wt.%** | **Thermal conductivity improved compared to pure PCM** | **Enhanced heat storage efficiency** |
| **3** | **Babapoor et al. (2022)** | **Paraffin + Graphene + Metal nanoparticles** | **Not specified** | **Increased thermal conductivity due to hybrid nano-additives** | **Faster solidification and improved thermal performance** |
| **4** | **Pereira et al. (2023)** | **Nano-enhanced PCM with graphene and metal nanoparticles** | **Various wt.%** | **Significant improvement in thermal conductivity** | **Improved energy storage and thermal cycling behavior** |
| **5** | **Mettawee & Assassa (2007)** | **PCM with conductive additives** | **Not specified** | **Enhanced heat transfer characteristics** | **Faster melting and solidification rates** |

From the literature survey, it is observed that although several studies have investigated nano-enhanced PCMs using graphene or metal nanoparticles individually, very limited research has explored the combined effect of copper nanoparticles and graphene nanoparticles together in paraffin wax [1]–[4]

**Applications of Phase Change Materials**

Phase Change Materials (PCMs) are widely used in various engineering and energy-related applications due to their ability to store and release large amounts of latent heat during phase transitions. Their capability to maintain nearly constant temperature during melting and solidification makes them highly suitable for thermal energy storage and temperature regulation systems.

One of the major applications of PCMs is in **solar thermal energy storage systems**, where they store excess heat during the daytime and release it during nighttime, thereby improving the efficiency and reliability of solar energy systems.

PCMs are also used in **building thermal management**, where they are incorporated into walls, ceilings, and roofing materials to regulate indoor temperature and reduce energy consumption for heating and cooling.

Another important application is in **electronics cooling**, where PCMs help dissipate heat generated by electronic components such as processors, batteries, and power modules. By absorbing excess heat during operation, PCMs prevent overheating and improve the lifespan and reliability of electronic devices.

PCMs are further used in **cold chain transportation and medical storage systems**, where they maintain stable temperatures for the storage and transportation of temperature-sensitive products such as vaccines, pharmaceuticals, and food items.

In addition, PCMs are applied in **thermal management of batteries and electric vehicles**, **textiles for thermal comfort**, and **industrial waste heat recovery systems**. With the development of nano-enhanced PCMs, their efficiency in heat transfer and energy storage has improved significantly, expanding their potential applications in modern thermal energy systems.

## Materials

The materials used in this study were selected based on their thermal properties, availability, and suitability for developing a nano-enhanced Phase Change Material (NEPCM). The composite PCM was prepared using paraffin wax as the base material and copper and graphene nanoparticles as thermal conductivity enhancers.

**Paraffin Wax (PW):**

Paraffin wax was used as the base Phase Change Material due to its high latent heat storage capacity, chemical stability, low cost, and availability. It also exhibits negligible supercooling and excellent thermal cycling stability, making it suitable for repeated melting and solidification processes. However, paraffin wax has relatively low thermal conductivity, typically in the range of **0.2–0.4 W/mK**, which limits its heat transfer performance in practical thermal energy storage systems. [1]

**Copper Nanoparticles (Cu-NPs):**

Copper nanoparticles were used as a metallic additive to enhance the thermal conductivity of the PCM. Copper possesses high thermal conductivity and can form efficient heat transfer pathways within the paraffin matrix. In this study, **10 wt.% copper nanoparticles** were added to all composite samples based on previous research that demonstrated significant thermal enhancement with this concentration. [2]

**Graphene Nanoparticles (GNPs):**

Graphene nanoparticles were selected due to their exceptionally high thermal conductivity, reported to be as high as **3000–5000 W/mK**. Graphene also offers a large surface area, excellent mechanical strength, and chemical stability, which contribute to improved thermal performance and structural stability of the composite PCM. To investigate the influence of graphene concentration on thermal conductivity, three different weight percentages were used: **1 wt.%, 2 wt.%, and 3 wt.%**. [3]

The combination of copper and graphene nanoparticles is expected to create an effective heat conduction network within the paraffin matrix, thereby significantly improving the thermal conductivity and heat transfer characteristics of the composite PCM.

**Sample Preparation**

The preparation of nano-enhanced Phase Change Material (NEPCM) samples was carried out using a controlled melting and ultrasonication process to ensure uniform dispersion of nanoparticles within the paraffin wax matrix. The entire procedure involved accurate weighing of materials, mixing under controlled temperature, ultrasonication, and solidification.

Step 1: Melting of Base PCM

Paraffin wax was used as the base Phase Change Material. The paraffin wax was in granules and placed in a clean beaker. The beaker was then heated using a hot plate at a temperature range of **45-47°C**, which is above the melting point of paraffin wax. Continuous stirring was maintained to ensure uniform melting and to prevent local overheating of the material.

Step 2: Weighing of Nanoparticles

Copper nanoparticles and graphene nanoparticles were weighed using a **precision digital microbalance** to obtain the required weight percentages. In this study, the concentration of copper nanoparticles was kept constant at **10 wt.%** for all samples, while the graphene nanoparticle concentration was varied to study its effect on thermal conductivity. Three different graphene concentrations were selected: **1 wt.%, 2 wt.%, and 3 wt.%**.

Step 3: Mixing of Nanoparticles with PCM

After the paraffin wax completely melted, the measured copper nanoparticles and graphene nanoparticles were slowly added into the molten paraffin wax. The mixture was continuously stirred to allow initial dispersion of nanoparticles throughout the PCM.

Step 4: Ultrasonication Process

To achieve better dispersion and to break down nanoparticle agglomerations, the mixture was subjected to **ultrasonication**. The molten PCM–nanoparticle mixture was placed in an ultrasonication bath and sonicated for approximately **4–5 hours**. During this process, the temperature was maintained to keep the paraffin wax in liquid form. Ultrasonication helps in distributing nanoparticles uniformly throughout the PCM and improves the stability of the composite material.

Step 5: Solidification of Composite PCM

After ultrasonication, the prepared nano-enhanced PCM mixture was poured into moulds or sample containers. The samples were allowed to cool naturally at **room temperature** until complete solidification occurred. Each sample was labelled properly based on its graphene concentration for further testing and analysis.

**Prepared Samples**

Three different composite PCM samples were prepared for experimentation:

| Cu Nanoparticle(g) | Graphene Powder(g) | Paraffin Wax(g) |
|---|---|---|
| 1. 2.0 (10%) | 0.2 (1%) | 17.8 |
| 2. 2.0 (10%) | 0.4 (2%) | 17.6 |
| 3. 2.0 (10%) | 0.6 (3%) | 17.4 |
| Total:6.0 (g) | Total:1.2 (g) | Total:52.8 (g) |

**Labels of Composite PCMs**

After the preparation and solidification of the nano-enhanced Phase Change Materials, each sample was labelled according to its nanoparticle composition to ensure proper identification during testing and analysis. The labels represent the percentage of copper nanoparticles and graphene nanoparticles added to the paraffin wax base PCM.

| Sample Label | Composition of PCM | Description |
|---|---|---|
| PW | Pure Paraffin Wax | Base PCM used for comparison |
| PW + Cu10 + G1 | Paraffin Wax + 10 wt.% Copper + 1 wt.% Graphene | Composite PCM Sample 1 |
| PW + Cu10 + G2 | Paraffin Wax + 10 wt.% Copper + 2 wt.% Graphene | Composite PCM Sample 2 |
| PW + Cu10 + G3 | Paraffin Wax + 10 wt.% Copper + 3 wt.% Graphene | Composite PCM Sample 3 |

### Scanning Electron Microscopy (SEM) Analysis

Scanning Electron Microscopy (SEM) was performed to examine the surface morphology and nanoparticle dispersion within the prepared nano-enhanced Phase Change Material (PCM) samples. SEM is an advanced characterization technique that uses a focused beam of electrons to obtain high-resolution images of material surfaces. This technique helps in identifying the distribution of nanoparticles, surface structure, particle clustering, and interaction between the base PCM and the added nanoparticles.

The SEM images of composites were taken by HITACHI S-3700N.

The SEM analysis was carried out using a **Field Emission Scanning Electron Microscope (FE-SEM)** operating at an accelerating voltage of approximately **15–30 kV** with magnifications around **20,000× – 45,000×**. The samples were examined at micro-scale ranges of approximately **3–5 µm**, allowing detailed observation of the dispersion of copper and graphene nanoparticles within the paraffin wax matrix.

**SEM Analysis of PCM with 1 wt.% Graphene (PW + Cu10 + G1)**

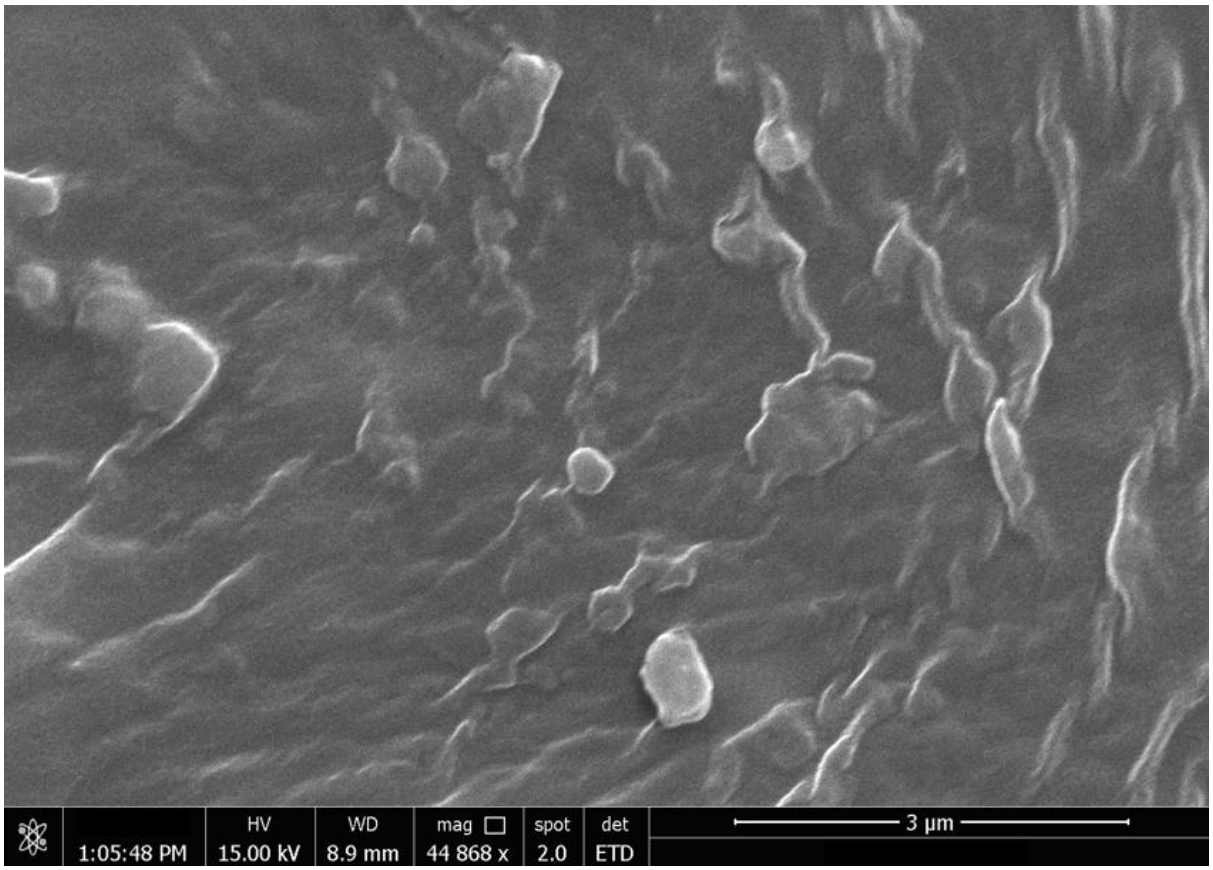



**Fig. 1 SEM micrograph of 87PW + 10Cu + 3Graphene composite PCM.**

**Figure 1** show the SEM images of the composite PCM containing **10 wt.% copper nanoparticles and 1 wt.% graphene nanoparticles**.

The micrographs reveal a relatively smooth paraffin matrix with small bright spots representing copper nanoparticles embedded within the PCM structure. The graphene sheets appear as thin irregular flakes distributed across the matrix surface. The nanoparticles are moderately dispersed throughout the paraffin wax, although small clusters can be observed in some regions.

The dispersion of nanoparticles indicates that the **ultrasonication process helped distribute the particles within the PCM**, creating potential conductive pathways for heat transfer. However, at lower graphene concentration, the network formation between nanoparticles appears limited.

**SEM Analysis of PCM with 2 wt.% Graphene (PW + Cu10 + G2)**

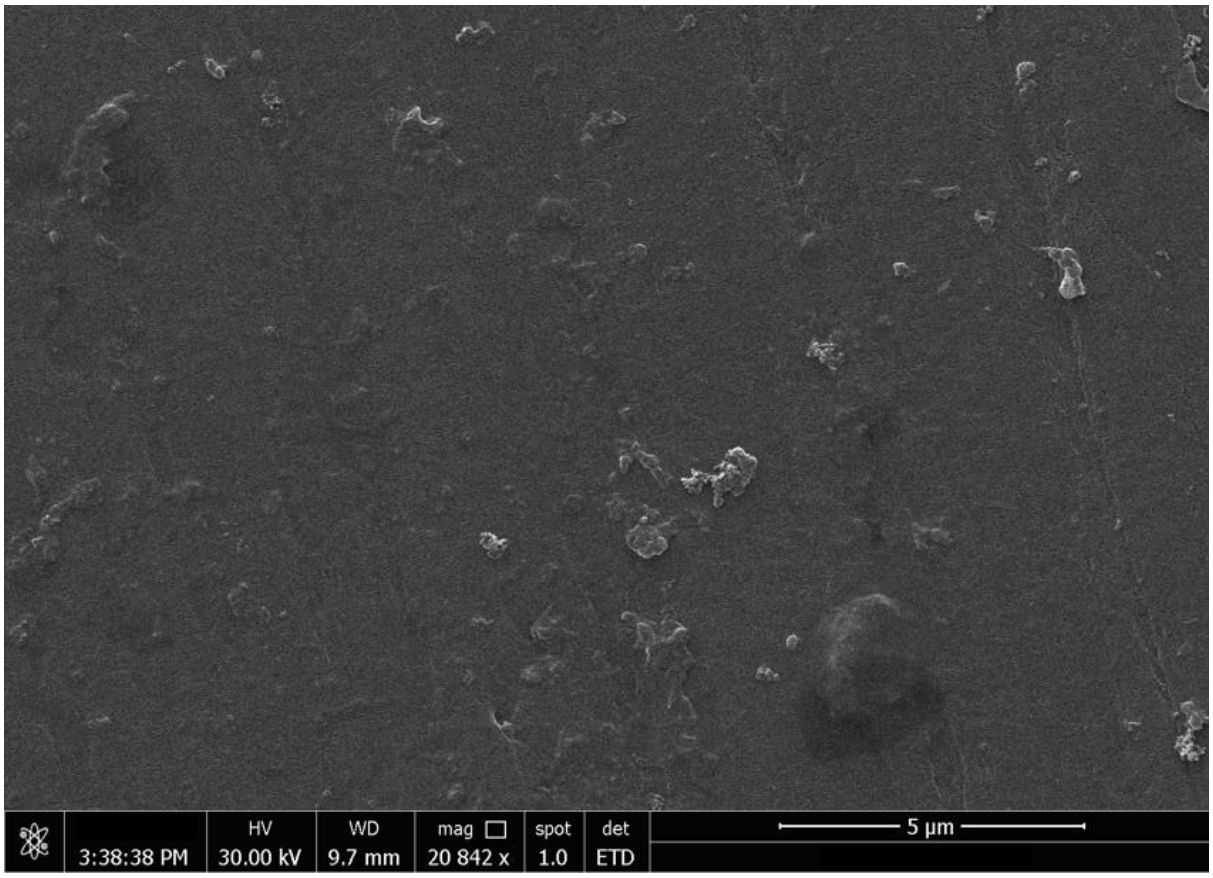



**Fig. 2 SEM micrograph of 88PW + 10Cu + 2Graphene composite PCM.**

**Figure 2** represent SEM images of the composite PCM containing **10 wt.% copper nanoparticles and 2 wt.% graphene nanoparticles**.

Compared to the 1% graphene sample, the micrographs show a **higher density of nanoparticle distribution** across the paraffin matrix. Graphene sheets appear more frequently and are spread over a larger surface area. Copper nanoparticles are visible as small bright particles dispersed across the structure.

The images suggest improved connectivity between graphene sheets and copper nanoparticles, forming **enhanced conductive pathways within the PCM matrix**. The dispersion appears relatively uniform with minimal agglomeration, indicating that the selected nanoparticle concentration and ultrasonication process effectively improved particle distribution.

**SEM Analysis of PCM with 3 wt.% Graphene (PW + Cu10 + G3)**

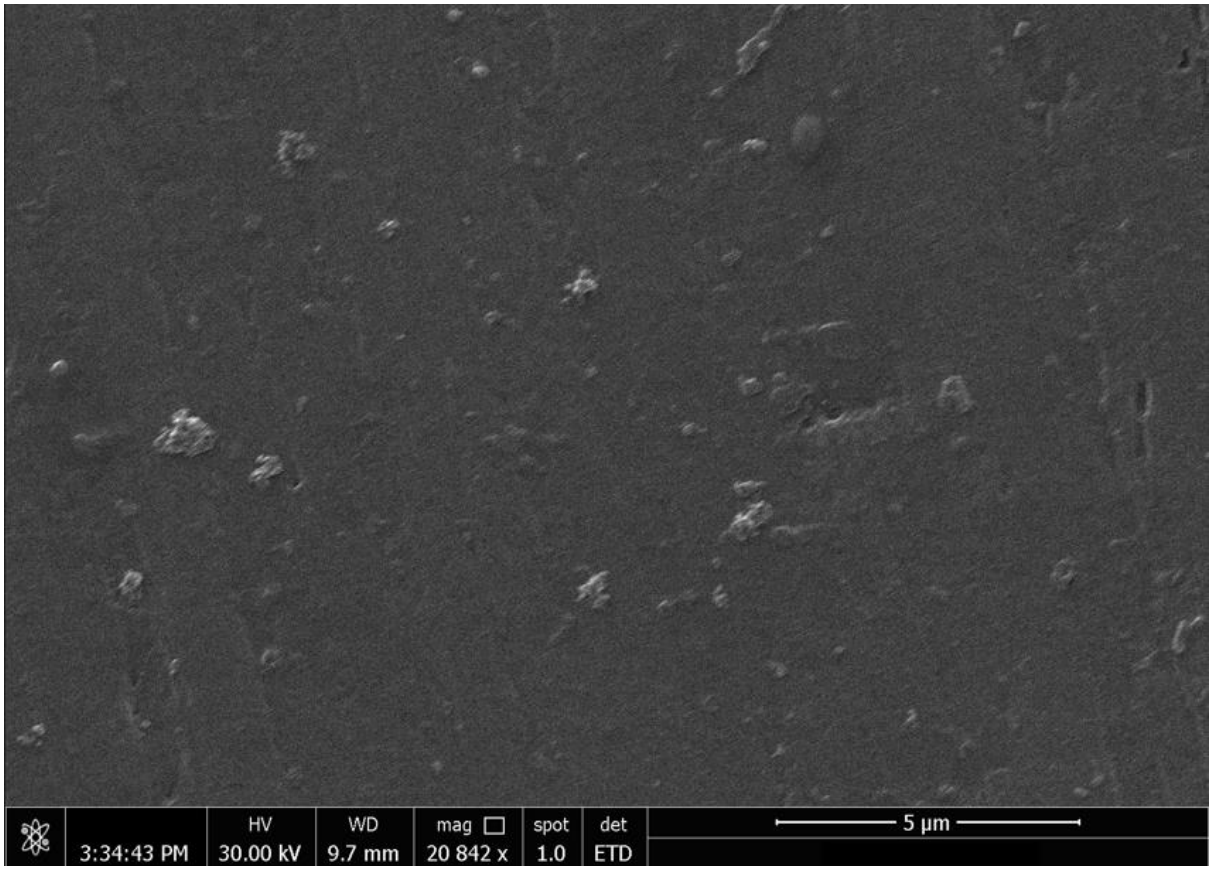



**Fig. 3 SEM micrograph of 89PW + 10Cu + 1Graphene composite PCM.**

**Figure 3** show SEM images of the PCM sample containing **10 wt.% copper nanoparticles and 3 wt.% graphene nanoparticles**.

The micrographs display a **higher concentration of graphene nanosheets distributed across the paraffin surface**. The graphene appears as layered or flake-like structures

embedded within the matrix. Copper nanoparticles are also visible within the composite structure.

Due to the increased graphene content, the nanoparticle network appears denser compared to the other samples. This dense distribution can contribute to **improved heat conduction pathways within the PCM**, which may significantly enhance thermal conductivity. In some regions, slight nanoparticle clustering can be observed, which is common at higher nanoparticle concentrations.

Overall SEM Observations

From the SEM analysis of all samples, the following observations can be made:

- The **ultrasonication technique successfully dispersed copper and graphene nanoparticles** within the paraffin wax matrix.
- Copper nanoparticles appear as bright particles embedded within the PCM structure.
- Graphene nanoparticles appear as thin sheet-like or flake-like structures.
- Increasing graphene concentration results in **higher nanoparticle density and improved conductive network formation**.
- The composite PCM samples show relatively **uniform dispersion with minimal agglomeration**, which is important for improving thermal conductivity.

These microstructural observations confirm the successful preparation of the nano-enhanced PCM and support the expectation that the addition of copper and graphene nanoparticles will enhance the thermal performance of paraffin wax.

## Thermal conductivity

### Instrument

The thermal conductivity of the prepared PCM samples was measured using a **FOX 50 Heat Flow Meter manufactured by TA Instruments**. This instrument is widely used for measuring the thermal conductivity of insulating materials and composite samples under controlled temperature conditions. The FOX 50 heat flow meter operates using two parallel plates that maintain a controlled temperature difference across the specimen.

### Range of Instrument

The FOX 50 heat flow meter is capable of measuring thermal conductivity over a wide range suitable for polymeric and composite materials. The instrument is designed to operate within a temperature range that allows accurate testing of materials such as paraffin-based PCMs. The measurement range is suitable for materials with relatively low to moderate thermal conductivity values.

**Accuracy of Measurement**

The instrument provides high precision and repeatability in thermal conductivity measurements. The FOX 50 heat flow meter is designed with calibrated heat flux sensors that ensure reliable measurement of heat transfer through the specimen under steady-state conditions.

**Test Method**

The thermal conductivity measurement is based on the **Heat Flow Meter Method**, which follows the principle of **one-dimensional steady-state heat transfer**. During the test, the specimen is placed between two parallel plates maintained at different temperatures. Heat flows from the hot plate to the cold plate through the specimen.

The thermal conductivity of the material is determined using **Fourier's law of heat conduction**, which relates the heat flux passing through the material to the temperature gradient across the sample thickness. [1]

**Specimen Size**

For the thermal conductivity test, the PCM samples were prepared in solid form and cut into specimens of uniform thickness to ensure proper contact between the specimen and the measurement plates. The samples were prepared with a thickness of approximately **5 mm** and were placed carefully between the plates to ensure uniform heat flow during the measurement process.

This testing method provides reliable thermal conductivity data for comparing the heat transfer performance of **pure paraffin wax and the developed nano-enhanced PCM samples containing copper and graphene nanoparticles**.

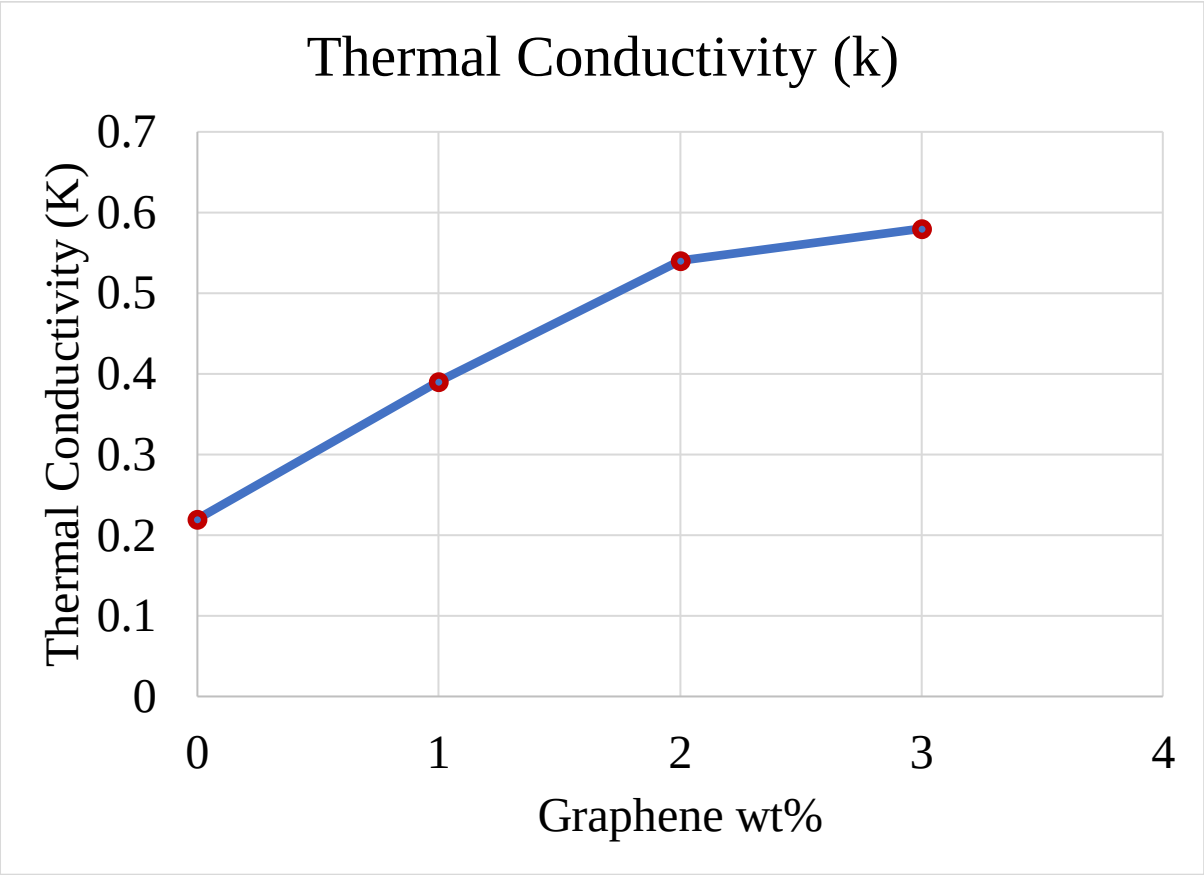
**Fig. 1 FOX 50 Heat Flow Meter**

| Composites | K values at 25 C | Enhacment % |
|---|---|---|
| Paraffin Wax | 0.22 | 0% |
| PWCuG1 | 0.39 | 77.27% |
| PWCuG2 | 0.54 | 145.45% |
| PWCuG3 | 0.58 | 163.63% |

**Graph 1: Comparison of Thermal Conductivity of PCM Samples**

Graph 1 shows the comparison between pure paraffin wax and the nano-enhanced PCM samples. The graph illustrates the increase in thermal conductivity after the addition of copper and graphene nanoparticles.

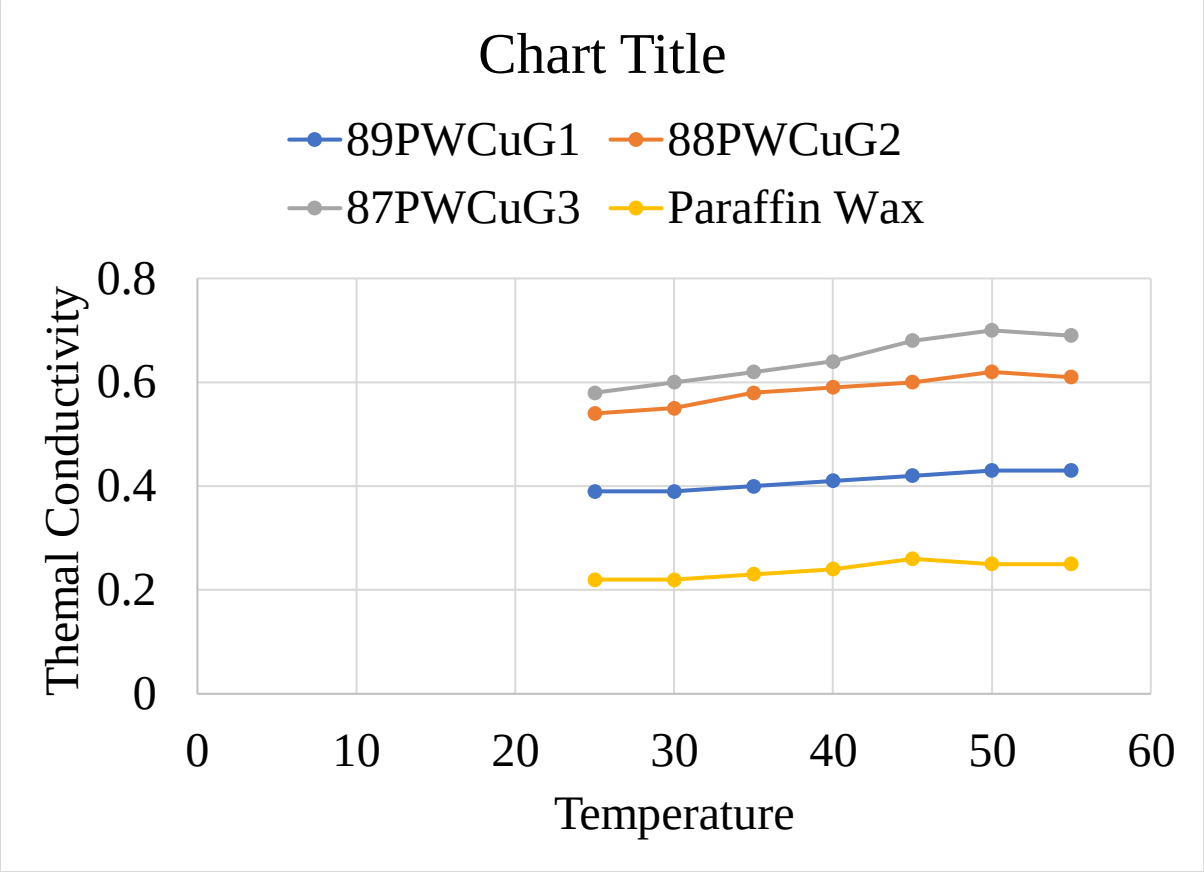
The results demonstrate that the addition of nanoparticles significantly improves the heat transfer capability of paraffin wax. Copper nanoparticles create conductive pathways, while graphene sheets form a thermal network within the PCM matrix, resulting in improved heat conduction.




**Graph 2: Effect of Graphene Concentration on Thermal Conductivity**

Graph 2 represents the effect of increasing graphene nanoparticle concentration on the thermal conductivity of the composite PCM.

From the graph, it can be observed that thermal conductivity increases with the increase in graphene content from **1 wt.% to 3 wt.%**. This improvement occurs because graphene possesses extremely high thermal conductivity and forms interconnected conductive networks inside the paraffin matrix.




**Result Analysis**

The experimental results obtained from SEM analysis and thermal conductivity measurements were analyzed to evaluate the effect of copper nanoparticles and graphene nanoparticles on the thermal performance of paraffin wax.

**SEM Analysis Results**

The SEM images confirmed the successful incorporation of copper and graphene nanoparticles into the paraffin wax matrix. The micrographs showed the presence of bright copper nanoparticles and thin graphene sheet-like structures distributed within the PCM structure. The ultrasonication process helped in achieving relatively uniform dispersion of nanoparticles throughout the paraffin matrix.

For the sample containing **1 wt.% graphene**, the nanoparticles were moderately distributed within the PCM structure, with limited formation of conductive networks. In the sample containing **2 wt.% graphene**, the distribution of nanoparticles appeared more uniform, and the graphene sheets were more visible within the matrix. The sample containing **3 wt.% graphene** showed a higher density of nanoparticles and a more interconnected structure, which indicates the formation of better heat transfer pathways within the composite PCM.

Overall, the SEM analysis confirmed that the addition of copper and graphene nanoparticles resulted in the formation of a nano-enhanced PCM structure with improved internal connectivity.

**Thermal Conductivity Analysis**

Thermal conductivity measurements revealed a significant improvement in the heat transfer capability of the composite PCM samples compared to pure paraffin wax. Pure paraffin wax exhibited the lowest thermal conductivity values due to its inherently poor heat transfer properties.

With the addition of **10 wt.% copper nanoparticles and increasing graphene concentrations**, the thermal conductivity values increased progressively. The presence of copper nanoparticles provided metallic heat conduction paths, while graphene nanoparticles formed thermally conductive networks inside the PCM matrix.

Among the prepared samples, the composite PCM containing **87PW + 10Cu + 3Graphene** showed the highest thermal conductivity values at all tested temperatures. This improvement indicates that increasing graphene concentration enhances the formation of conductive pathways, which accelerates heat transfer within the material.

**Graphical Analysis**

The graphical representation of the experimental results clearly shows the relationship between thermal conductivity, temperature, and nanoparticle concentration. The graph of **thermal conductivity versus temperature** indicates a gradual increase in thermal conductivity with rising temperature for all samples.

Similarly, the graph of **thermal conductivity versus graphene weight percentage** demonstrates that thermal conductivity increases as the graphene concentration increases from **1 wt.% to 3 wt.%**. This trend confirms that graphene nanoparticles play a significant role in enhancing heat transfer within the PCM.

**LATENT HEAT**

Latent heat is the amount of heat energy absorbed or released by a material during a phase change process without any change in temperature. In Phase Change Materials (PCMs), latent heat plays a crucial role in thermal energy storage because it allows the material to store a large amount of heat during melting and release it during solidification at a nearly constant temperature. [1]

When paraffin wax is heated, it absorbs heat energy and undergoes a phase transition from solid to liquid. During this process, the temperature of the material remains almost constant,

and the absorbed energy is stored in the form of latent heat. Similarly, during cooling, the PCM releases the stored latent heat while changing from liquid back to solid.

The latent heat storage capacity of a PCM is an important parameter in determining its effectiveness in thermal energy storage applications. Materials with high latent heat can store and release more energy, making them more efficient for applications such as solar thermal systems, electronic cooling, and thermal management systems.

In nano-enhanced PCMs (NEPCMs), the addition of nanoparticles such as copper and graphene primarily improves thermal conductivity. However, it may slightly affect the latent heat capacity depending on the concentration and distribution of nanoparticles. Generally, a small reduction in latent heat may occur due to the replacement of base PCM by nanoparticles, but the overall heat transfer performance improves significantly due to enhanced thermal conductivity.

Therefore, an ideal PCM should have high latent heat capacity along with improved thermal conductivity to achieve efficient thermal energy storage and heat transfer performance.

| **Materials** | **Latent Heat** | **Enhacment %** |
|---|---|---|
| **Paraffin Wax** | **249 KJ/kg** | **0%** |
| **PWCuG1** | **276** | **10.84%** |
| **PWCuG2** | **260** | **4.10%** |
| **PWCuG3** | **249** | **0.00%** |

**Latent Heat Table**

**Overall Performance**

The results demonstrate that the hybrid combination of **copper nanoparticles and graphene nanoparticles** significantly improves the thermal conductivity of paraffin wax. The composite PCM with **3 wt.% graphene** exhibited the best performance due to the formation of a dense conductive network within the paraffin matrix.

Therefore, the developed nano-enhanced PCM shows strong potential for applications in **thermal energy storage, electronic cooling, and thermal management systems**, where improved heat transfer characteristics are required.

**Conclusion**

1. A nano-enhanced Phase Change Material (PCM) was successfully developed by incorporating **copper nanoparticles and graphene nanoparticles into paraffin wax**.
2. **Paraffin wax** was selected as the base PCM due to its **high latent heat storage capacity, chemical stability, and suitability for thermal energy storage applications**.
3. **Copper nanoparticles and graphene nanoparticles** were used as thermal conductivity enhancers to improve the heat transfer capability of paraffin wax.
4. Three composite PCM samples were prepared using **10 wt.% copper nanoparticles and varying graphene concentrations of 1 wt.%, 2 wt.%, and 3 wt.%**.
5. The **ultrasonication method** was used for sample preparation, which ensured **uniform dispersion of nanoparticles and reduced agglomeration within the PCM matrix**. [5]
6. **SEM analysis** confirmed the successful incorporation and distribution of copper and graphene nanoparticles within the paraffin matrix.
7. The microstructural observations indicated that **increasing graphene concentration improved the connectivity between nanoparticles**, which helped create better heat transfer pathways.
8. **Thermal conductivity testing** showed a significant improvement in the heat transfer performance of the nano-enhanced PCM compared to pure paraffin wax.
9. Among all the prepared samples, the composite PCM containing **87PW + 10Cu + 3Graphene** exhibited the **highest thermal conductivity values**.
10. The results demonstrate that the **combined addition of copper and graphene nanoparticles effectively enhances the thermal conductivity of paraffin wax**.
11. The developed nano-enhanced PCM has strong potential for applications in **thermal energy storage systems, electronics cooling, solar thermal systems, and other thermal management technologies**.
12. Overall, the study confirms that **hybrid nanoparticle reinforcement is an effective method to overcome the low thermal conductivity limitation of conventional PCMs**.

**Future Scope**

The present study focuses on improving the thermal conductivity of paraffin wax by incorporating copper and graphene nanoparticles. Although the results show significant improvement in thermal performance, further research can be carried out to expand the application and effectiveness of nano-enhanced Phase Change Materials (NEPCMs).

1. **Differential Scanning Calorimetry (DSC) Analysis:**
   Future work can include DSC testing to determine the **melting temperature, solidification temperature, and latent heat of fusion** of the developed nano-enhanced

PCM samples. This analysis will help evaluate how the addition of copper and graphene nanoparticles influences the phase change behaviour and energy storage capacity of the material.

2. **Thermogravimetric Analysis (TGA):**
   TGA can be performed to study the **thermal stability and decomposition characteristics** of the composite PCM at different temperature ranges. This test will provide important information about the thermal durability of the nano-enhanced PCM under high-temperature operating conditions.
3. **Thermal Cycling Stability:**
   Long-term **thermal cycling tests** can be conducted to evaluate the reliability and stability of the PCM during repeated melting and solidification cycles. This will help determine whether the composite PCM maintains its thermal properties after many heating and cooling cycles.
4. **Prediction Using Machine Learning Models:**
   Advanced computational techniques such as **Machine Learning (ML)** can be used to predict the thermal behaviour of nano-enhanced PCMs. By training models using experimental data, it may be possible to predict how parameters such as **thermal conductivity (k), latent heat, and melting temperature** change with different nanoparticle concentrations, materials, and preparation conditions. This approach can significantly reduce experimental time and help identify optimal material compositions.
5. **Development of Advanced Nano-Enhanced PCMs:**
   Future research may explore other **hybrid nanoparticle combinations or different base PCMs** to further improve thermal conductivity and energy storage performance for advanced thermal management applications.

Overall, further research in these areas can help in developing more efficient nano-enhanced PCMs with improved thermal performance and broader industrial applications.

**Nomenclature**

| Symbol / Abbreviation | Description | Unit |
|---|---|---|
| PCM | Phase Change Material | — |
| NEPCM | Nano-Enhanced Phase Change Material | — |
| PW | Paraffin Wax | — |
| Cu | Copper Nanoparticles | — |
| GNP | Graphene Nanoparticles | — |
| wt.% | Weight Percentage | % |
| k | Thermal Conductivity | W/m·K |
| T | Temperature | °C |
| SEM | Scanning Electron Microscopy | — |
| FE-SEM | Field Emission Scanning Electron Microscopy | — |
| Cu-NPs | Copper Nanoparticles | — |
| GNPs | Graphene Nanoparticles | — |

## References


1. Mettawee, E., and Assassa, G., "Experimental study of enhanced heat transfer of phase change materials in latent heat storage systems," Energy Conversion and Management, 2007.
2. Babapoor, A., Karimi, N., and Afrand, M., "Enhancement of paraffin phase change material using graphene and metal oxide nanoparticles," Renewable Energy, 2022.
3. Pereira, D., Silva, F., and Costa, J., "Nano-enhanced phase change materials for energy harvesting and thermal management applications: A review," Energy Conversion and Management, 2023.
4. Chang, X., Li, Y., and Zhang, H., "Form-stable copper–graphene composite phase change materials for enhanced solar thermal energy storage," Journal of Energy Storage, 2023.
5. Calotă, A., Popescu, M., and Marinescu, M., "Hybrid nanoparticle-enhanced phase change materials for improved thermal energy storage performance," Applied Thermal Engineering, 2024.